

## **ULIP Integration Requirement Document**

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### Revision History

| Version | Date       | Author         | Reviewer     | Significant Changes |
|---------|------------|----------------|--------------|---------------------|
| 1.0     | 20/06/2023 | Ayush Arya     | Nayab Ali    | Baseline Version    |
| 2.0     | 01/10/2024 | Ayush Arya     | Nayab Ali    | Updated             |
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### Introduction

#### 1.1 Purpose

The purpose of this document is to define and provide details for accessing API for EWAY BILL system information (information about ewbNo available in EWAY Bill system) from ULIP system.

This document details the following points:

1. **Data Integration between EWAYBILL and ULIP web service:** Identification of operational points where data integration between **EWAYBILL** and ULIP web service will be needed.
2. **Technical Approach** for performing Data Integration

### The Overall Description

User will share the information such as **EWB number** based on data send by user ULIP system will call respective **EWAYBILL** API for accessing information.

Following are the point for accessing vehicle information: -

1. This API will share data based on information provided by user.
2. ULIP will expose API for getting data from EWAYBILL API.
3. **EWAYBILL/01** API will take EWB number from user and get data from EWAYBILL API.

#### 1.2 Data Integration between EWAYBILL System and ULIP application

User will share information to ULIP system based on data shared ULIP will provide data to user.

#### 1.3 EWAYBILL/01

This API will take **EWB (e-way bill) number** and connect with EWAYBILL API to get data.

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| S. No. | Field Name   | Field Description                                     | Format    | Required/Not – Required | Length |
|--------|--------------|-------------------------------------------------------|-----------|-------------------------|--------|
| 1      | <b>ewbNo</b> | Enter the EWB number as available in EWAYBILL system. | [0-9]{12} | Required                | 12     |

### 1.3.1 Technical Approach

1. All the data will be shared through rest web services.
2. ULIP system will provide **EWB number** for which detail is required.
3. EWAYBILL API will share details of e-way bill information which exist in EWAYBILL system in JSON format.
4. EWAYBILL Integration will add JSON in **response** key in response body.
5. ULIP system will provide the data in JSON Object format in response body.

#### Example:

EWB number **101000609218** is shared by ULIP system, then EWAYBILL-integration will make a request to EWAYBILL API.

#### Request:

The request must be of the following format, where e-way bill (EWB) number is passed within the URL body in JSON format.

#### For staging -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/EWAYBILL/01>

```
{  
  "ewbNo": "101000609218"  
}
```

#### Curl URL

```
curl --location --  
request POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/EWAYBILL/01' \  
--  
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJlbGwIiwiaWF0IjoxNjg3MjM4MjU3LCJhcHBzIjoizGF0YXB1c2gifQ.auwp_xfX_FBUJLB-7iNOyr5WalH2PeboAok-0uAADvNVADPK1C-MEW9ne5FKttsuDjoOTdkbs_RC9xtMLrCF6g' \  
--header 'Content-Type: application/json' \  
--header 'Accept: application/json' \
```

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```
--data-raw '{  
  "ewbNo": "101000609218"  
}'
```

### For production -

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/EWAYBILL/01>

```
{  
  "ewbNo": "101000609218"  
}
```

### Curl URL

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/EWAYBILL/01' \  
--header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJlbGlwIiwiaWF0IjoxNzI3NzYyNzcxLCJhcHBzIjoiZGF0YXB1c2gifQ.9_kJwxuXfJYomT0QeMgOSWi01EaxXQkZQ1JLevfNo-vJ5vjCY9oIemZ_B4hiiq2ZCog-yAbIUgI6YN-TVie3cA' \  
--header 'Content-Type: application/json' \  
--header 'Accept: application/json' \  
--data '{  
  "ewbNo": "871347047092"  
}'
```

For obtaining authentication token, refer [Section 1.4](#)

### Response

To acknowledge the above request, ULIP will send the below response -

#### 1. Invalid ewbNo number

```
{  
  "response": null,  
  "error": "true",  
  "code": "400",  
  "message": "Data format failed OR wrong value entered at: ewbNo. Format should follow [0-9]{12}"  
}
```

#### 2. In case nhno number exist

```
{  
  "response": [  
    {  
      "response": {
```

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```
        "fromPincode": 301404,
        "hsnCode": "",
        "ewayBillDate": "29/11/2017 04:30:00 PM",
        "validUpto": " 11:59:00 PM",
        "ewbNo": 101000609218,
        "toPincode": 302014,
        "VehicleListDetails": [
            {
                "vehicleNo": "RJ14CG4508",
                "enteredDate": "29/11/2017 04:30:00 PM",
                "transMode": "1"
            }
        ],
        "status": "ACT"
    },
    "responseStatus": "SUCCESS"
}
],
"error": "false",
"code": "200",
"message": "Success"
}
```

### 3. In case nhno number not exist

```
{
  "response": [
    {
      "response": {
        "errorCodes": "404"
      },
      "responseStatus": "ERROR"
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

**Status Code 200:** EWAYBILL data response

**Status Code 400:** Bad request (Invalid json syntax, invalid json key)

**Status Code 401 or 403:** Unauthenticated or Unauthorized

**Status Code 500:** Some internal server error occurred

**Status Code 502:** Server is not responding.

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### 1.4 Data Transmission & Authentication Mechanism

All data exchange would be done over secure HTTP (HTTPS). Request from ULIP's system would be made in EWAYBILL system through a data exchange URL that has following form:

#### For staging -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/EWAYBILL/01>

```
{
  "ewbNo": "101000609218"
}
```

Access to above data exchange URL would be authorized by use of a security mechanism implemented by the ULIP system. It is as follows:

- First time access: It would require a username and password [which would be shared with user beforehand]. Once user hit with valid user id and password, ULIP system will return a basic authorization token to user, using that token user will communicate with ULIP system until token will not get expired.

Brief summary of HTTP Basic Authentication is as follows:

First time when User will hit ULIP secured API through username and password for access token as follow-

```
curl --location --request POST
'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/user/login' \
  --header 'Accept: application/json' \
  --header 'Content-Type: application/json' \
  --data-raw '{
    "username": "xxxx",
    "password": "xxxx@123"
  }'
```

#### Authorization:

##### Bearer

eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiI1bGlwliwiaWF0IjoxNjg3MjM4MjU3LCJhcHBzIjoiaZGF0YXB1c2gifQ.auwp\_xfX\_FBUJLB-7iNOyr5Wa1H2PeboAok-0uAADvNVADPK1C-MEW9ne5FKttsuDjoOTdkbs\_RC9xtMLrCF6g'

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Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired and user need to re authenticate.

Using The above token, user can request ULIP system for cargo details as follow -

```
curl --location --  
request POST 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/EWAYBILL/01' \  
--  
header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWliOiJ1bGlwliwiaWF0IjoxNjg3MjM4MjU3LCJhcHBzIjoizGF0YXB1c2gifQ.auwp_xfX_FBUJLB-7iNOyr5Wa1H2PeboAok-0uAADvNVADPK1C-MEW9ne5FKttsuDjoOTdkbs_RC9xtMLrCF6g' \  
--header 'Content-Type: application/json' \  
--header 'Accept: application/json' \  
--data-raw '{  
  "ewbNo": "101000609218"  
}'
```

For production –

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/user/login' \  
--header 'Accept: application/json' \  
--header 'Content-Type: application/json' \  
--data-raw '{  
  "username": "xxxx",  
  "password": "xxxx@123"  
}'
```

**Authorization:**

**Bearer**

**eyJhbGciOiJIUzUxMiJ9.eyJzdWliOiJ1bGlwliwiaWF0IjoxNzI3MTYxMDgwLCJhcHBzIjoizGF0YXB1c2gifQ.XhVZm7NiCD05-v9091WJQhYzdk1a0Shs\_Ley0T4R2MX63swRyxHbHxh6bVXtzqWvWiSLwwNGRQIcqGhr49hpGg**

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represent user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired and user need to re authenticate.

Using The above token, user can request ULIP system for details as follow -

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```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/EWAYBILL/01' \  
--header 'Authorization: Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGlwliwiaWF0IjoxNzI3NzYyNzcxLCJhcHBzIjoZGF0  
YXB1c2gifQ.9_kJwxuXfJYoMt0QeMgOSWi01EaxXQkZQ1JLevfNo-  
vJ5vjCY9olemZ_B4hiiq2ZCog-yAbIUgL6YN-TVie3cA' \  
--header 'Content-Type: application/json' \  
--header 'Accept: application/json' \  
--data '{  
  "ewbNo": "871347047092"  
}'
```